

Lab 8: AWS/VSCode Setup

GW CS 2541W: Database Systems and Team Projects - 2026

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AWS RDS Setup

Create a MySQL Database Server in the Cloud



Today's lab you will be actively following along with the slides on your computer

Today's lab is going to be different from all the all the other lab's from the rest of the semester

Setting up AWS RDS is very finicky and this lab will require you to follow along step by step exactly. One singular mess up at an earlier step can completely mess up your configuration and ruin your set up for the **final project**, leading to unexpected bugs and issues.

We strongly ask you to pay very close attention and raise you hand immediately if you are unsure about any specific step of this process that we will be walking you through

Log into AWS Academy and click **Modules** → **AWS Academy Learner Lab**

▼ AWS Academy Learner Lab



Launch AWS Academy Learner Lab

▼ AWS Academy Learner Lab Resources



Demo - How to Access Learner Lab



Demo - General Troubleshooting Tips

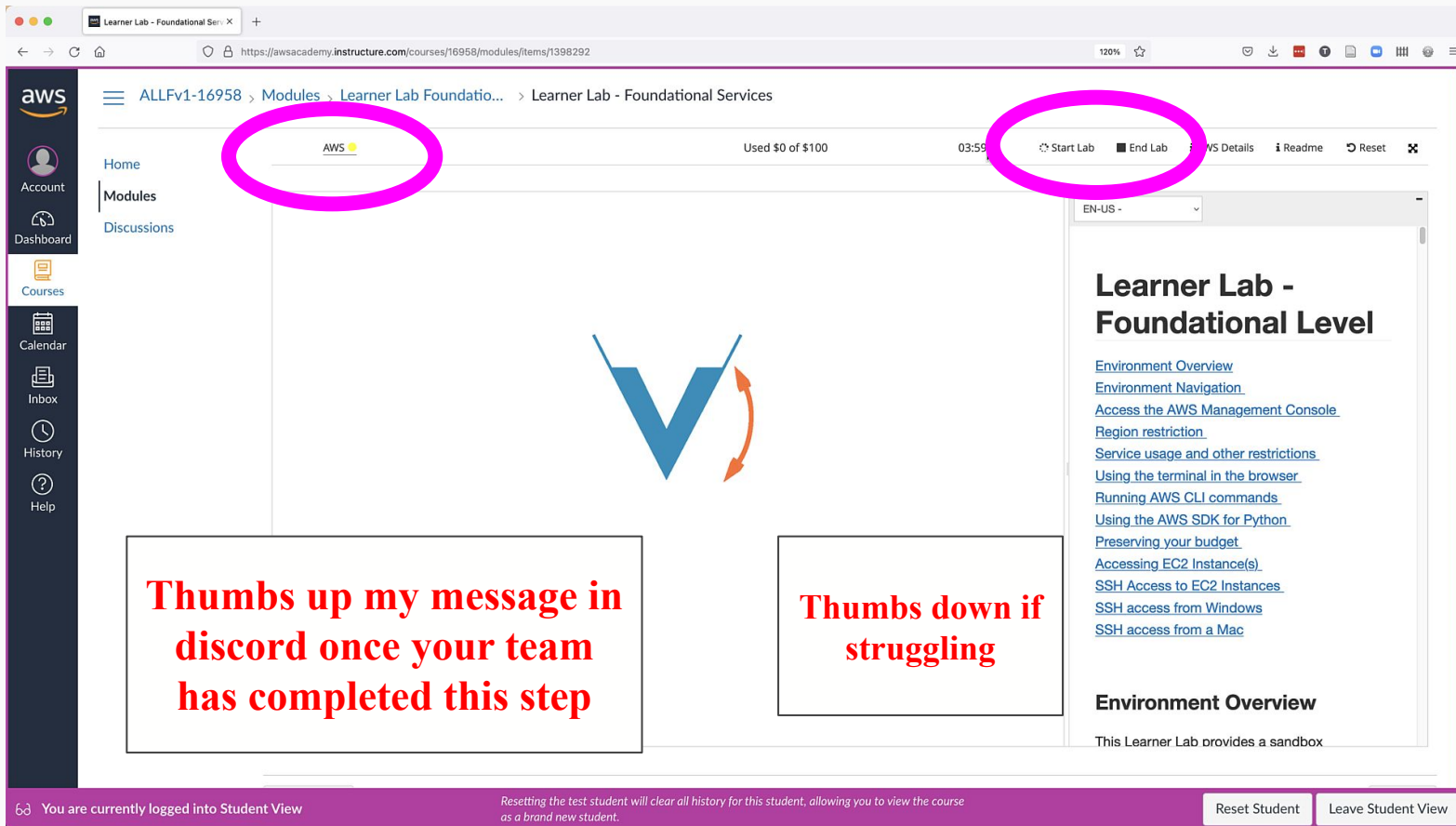


Demo - How to Launch Services through AWS Console



Learner Lab Activity - CodeWhisperer

Click **Start Lab**, wait, then click **AWS** once green



The screenshot shows the AWS Academy Learner Lab interface. The breadcrumb navigation at the top reads: **ALLFv1-16958** > Modules > Learner Lab Foundatio... > Learner Lab - Foundational Services. The **AWS** link in the breadcrumb is circled in pink. In the top right corner, the **Start Lab** button is circled in pink. A large blue 'V' with an orange curved arrow pointing to it is centered on the page. Below the 'V' are two white boxes with red text. On the right side, a sidebar contains a list of links under the heading 'Learner Lab - Foundational Level'. At the bottom of the page, a purple bar contains the text 'You are currently logged into Student View', a note about resetting the test student, and two buttons: 'Reset Student' and 'Leave Student View'.

Thumbs up my message in discord once your team has completed this step

Thumbs down if struggling

Learner Lab - Foundational Level

- [Environment Overview](#)
- [Environment Navigation](#)
- [Access the AWS Management Console](#)
- [Region restriction](#)
- [Service usage and other restrictions](#)
- [Using the terminal in the browser](#)
- [Running AWS CLI commands](#)
- [Using the AWS SDK for Python](#)
- [Preserving your budget](#)
- [Accessing EC2 Instance\(s\)](#)
- [SSH Access to EC2 Instances](#)
- [SSH access from Windows](#)
- [SSH access from a Mac](#)

Environment Overview

This Learner Lab provides a sandbox

63 You are currently logged into Student View

Resetting the test student will clear all history for this student, allowing you to view the course as a brand new student.

Reset Student Leave Student View

In AWS Management Console - Type "RDS" into the search bar, then click **Databases**

The screenshot shows the AWS Management Console interface. In the top left, the search bar contains the text "rds", which is circled in pink. Below the search bar, the search results for "rds" are displayed. The "Services" section is highlighted with a pink oval, and the "RDS" service is selected, also circled in pink. A red text box with a white border is overlaid on the search bar area, containing the text "Search 'rds' in the search bar on the top left!". The "Features" section is also visible, showing various AWS services and features. The right sidebar contains the "Console Home" section with links to "Getting started with AWS", "Training and certification", and "What's new with AWS?". The bottom of the page shows the "Build a solution" section and the "Trusted Advisor" section.

Search results for 'rds'

Services [See all 8 results](#)

- RDS** ☆
Managed Relational Database Service
- Top features**
[Dashboard](#) [Databases](#) [Query Editor](#) [Performance Insights](#) [Snapshots](#)
- AWS Glue DataBrew** ☆
Visual data preparation tool to clean and normalize data for analytics and machine le...
- Kinesis** ☆
Work with Real-Time Streaming Data
- Amazon OpenSearch Service** ☆
Run and Scale OpenSearch and Elasticsearch Clusters (successor to Amazon Elasticse...

Features [See all 25 results](#)

- Dashboards**
[CloudWatch feature](#)
- Security standards**
[Security Hub feature](#)
- Redis**
[ElastiCache feature](#)
- Users**
[IAM feature](#)

Blogs [See all 1,380 results](#)

Welcome to AWS

- Getting started with AWS** [Get started](#)
Learn the fundamentals and find valuable information to get the most out of AWS.
- Training and certification** [Get started](#)
Learn from AWS experts and advance your skills and knowledge.
- What's new with AWS?** [Get started](#)
Discover new AWS services, features, and Regions.

Console Home

Console Home displays widgets with important information about your AWS environment.

Customize your Console Home

You can add a widget from the actions menu, change its size from the widget menu, or drag and drop it to change its position. Changes you make do not affect others using your shared account.

Learn more [Get started](#)

[Getting started with the AWS Management Console](#)

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Click Create Database. Use these options when filling out the database configuration

The screenshot shows the AWS RDS Management Console 'Create Database' page. The 'Version' is set to 'MySQL 8.0.27'. Under 'Templates', the 'Free tier' option is selected and highlighted with a pink oval. Below, under 'Availability and durability', the 'Single DB instance' option is selected. In the 'Settings' section, the 'DB instance identifier' is set to 'database-1'. The 'Master username' is also visible.

Method: Standard Create

Engine Type: MySQL

Template: Free Tier

Instance Identifier: something like REGS26-LastName

Master Password: something secure that you write down!

Storage Autoscaling: Disable

Public Access: YES

Additional Configuration

Initial DB name: university

Disable Automatic Backups

Disable Auto minor version upgrade

Then click **Create Database**

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Thumbs down if struggling

In Database Status window, click the Security Group

The screenshot shows the Amazon RDS console for a database instance named 'instructor-1'. The instance is in the 'Creating' state. The 'Connectivity & security' tab is active, showing details for the endpoint, port, and security groups. A pink circle highlights the 'VPC security groups' link, which is currently set to 'default (sg-0384ac637a32371f)'. The console also shows the instance's CPU usage, current activity, and engine type (MySQL Community).

Summary

DB identifier	CPU	Status	Class
instructor-1	0.00%	Creating	db.t2.micro

Role	Current activity	Engine	Region & AZ
Instance	0 Connections	MySQL Community	-

Connectivity & security

Endpoint & port	Networking	Security
Endpoint -	Availability Zone -	VPC security groups default (sg-0384ac637a32371f) Active
Port -	VPC vpc-0b699f540a8be85f4	Yes
	Subnet group default-vpc-0b699f540a8be85f4	Certificate authority rds-ca-2019
	Subnets subnet-0da09d5a8f868f76c subnet-05d9291f53956f38b subnet-03954f8ea006d729e subnet-08c87f4b16adaf686 subnet-0a8720349a826bbab subnet-0d25c733e03dee097	Certificate authority date August 22, 2024, 01:08 (UTC+1:08)

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Add a new Inbound Rule - Port 3306 - Source IP 0.0.0.0/0

EC2 > Security Groups > sg-03846ac637a32371f - default > Edit inbound rules

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
sg-089996e2a73bcdbe6	All traffic	All	All	Custom	
-	Custom TCP	TCP	3306	Custom	

[Add rule](#)

[Cancel](#) [Preview changes](#) [Save rules](#)

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Check your DB creation status. When ready, copy host info

The screenshot displays the Amazon RDS Management Console for an instance named 'instructor-1'. The interface includes a left-hand navigation menu with options like Dashboard, Databases, Query Editor, and Performance insights. The main content area shows the instance's summary, connectivity & security details, and security group rules.

Summary

DB identifier instructor-1	CPU 0.00%	Status Available	Class db.t2.micro
Role Instance	Current activity 0 Connections	Engine MySQL Community	Region & AZ us-east-1c

Connectivity & security

Endpoint

Endpoint
instructor-1.ck8ualavedvt.us-east-1.rds.amazonaws.com

3306

Networking

Availability Zone
us-east-1c

VPC
[vpc-0b699f540a8be85f4](#)

Subnet group
default-vpc-0b699f540a8be85f4

Subnets
[subnet-0da09d5a8f868f76c](#)
[subnet-05d9291f53956f38b](#)
[subnet-03954f8ea006d729e](#)
[subnet-08c87fab16ada6f86](#)
[subnet-0a8720349a826bbab](#)
[subnet-0d25c733e03dee097](#)

Security

VPC security groups
[default \(sg-03846ac57a32371f\)](#)
Active

Public accessibility
Yes

Certificate authority
rds-ca-2019

Certificate authority date
August 22, 2024, 01:08 (UTC+1:08)

Security group rules (3)

Filter by security group rules

Test from AWS Academy Lab Console (other tab)

The screenshot shows the AWS Academy Lab Console interface. The browser tab is labeled 'Learner Lab - Foundational Services'. The URL is <https://awsacademy.instructure.com/courses/16958/modules/items/1398292>. The left sidebar contains navigation links: Home, Modules, Discussions, Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area displays a terminal session with the following output:

```
ddd_v1_w_9M5_1123374@runweb5146:~$ ls
ddd_v1_w_9M5_1123374@runweb5146:~$ mysql -h instructor-1.ck8ualavedvt.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Copyright 2000, 2021, Oracle and/or its affiliates.
Your MySQL connection id is 16
Server version: 8.0.27 Source distribution

Copyright (c) 2000, 2021, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql       |
| performance_schema |
| sys        |
| university  |
+-----+
5 rows in set (0.07 sec)

mysql> use university;
Database changed
mysql> show tables;
Empty set (0.07 sec)

mysql>
```

Commands:

**mysql -h [endpoint] -u
admin -p**

Enter password

**mysql> show
databases;**

mysql> use university;

mysql> show tables;

**Thumbs up my message in
discord once your team**

Thumbs down if

MySQL vs SQLite

Code / System Differences



SQLite vs MySQL

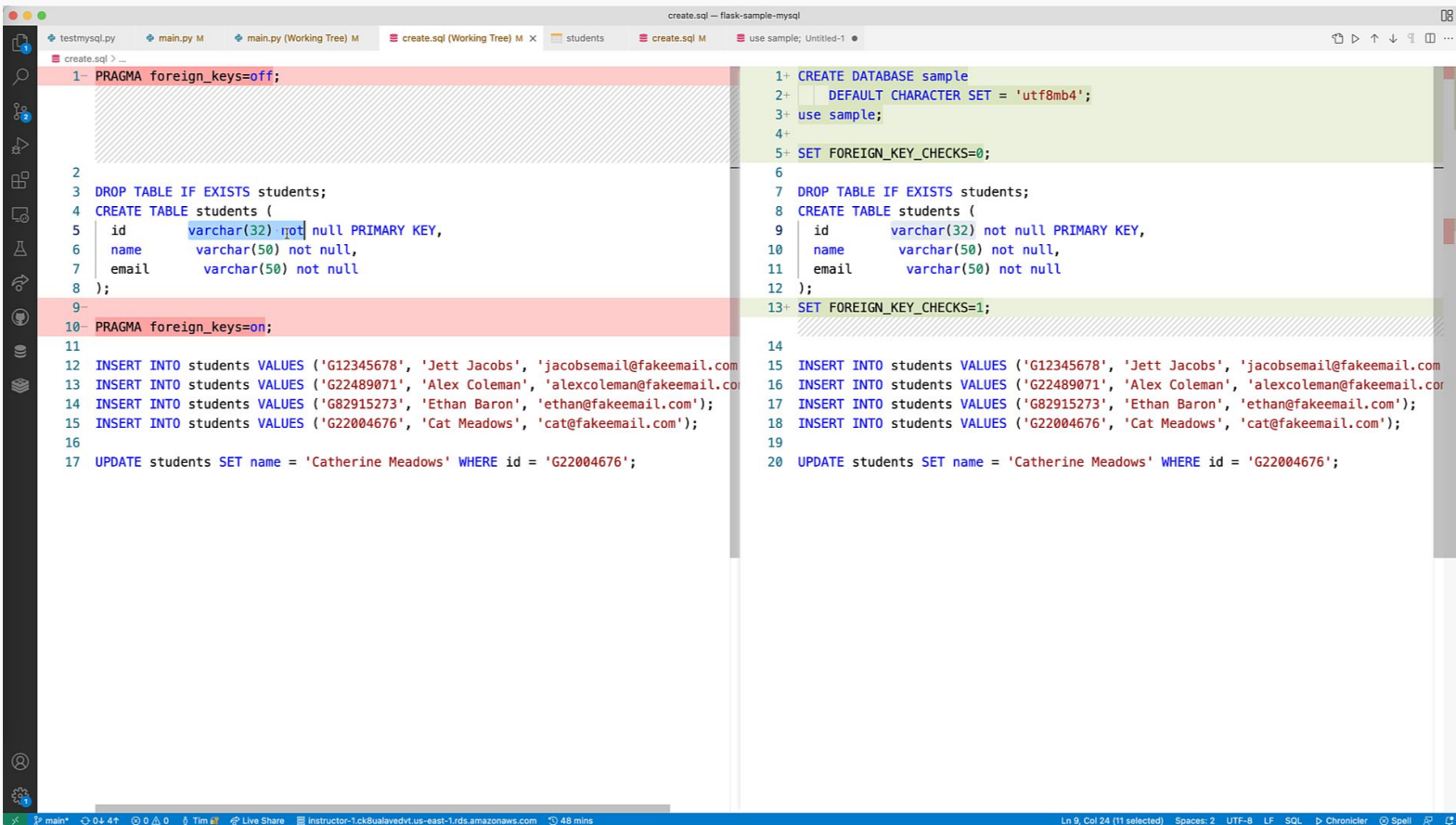
SQLite: embedded database

- Simple, single file based DB integrated in your software
- Great for mobile apps, or single user applications

MySQL: full featured database server

- Independent server application
- Can be deployed across a cluster of servers
- Supports larger scale and stronger reliability guarantees

SQL Code Differences



```
create.sql -- flask-sample-mysql

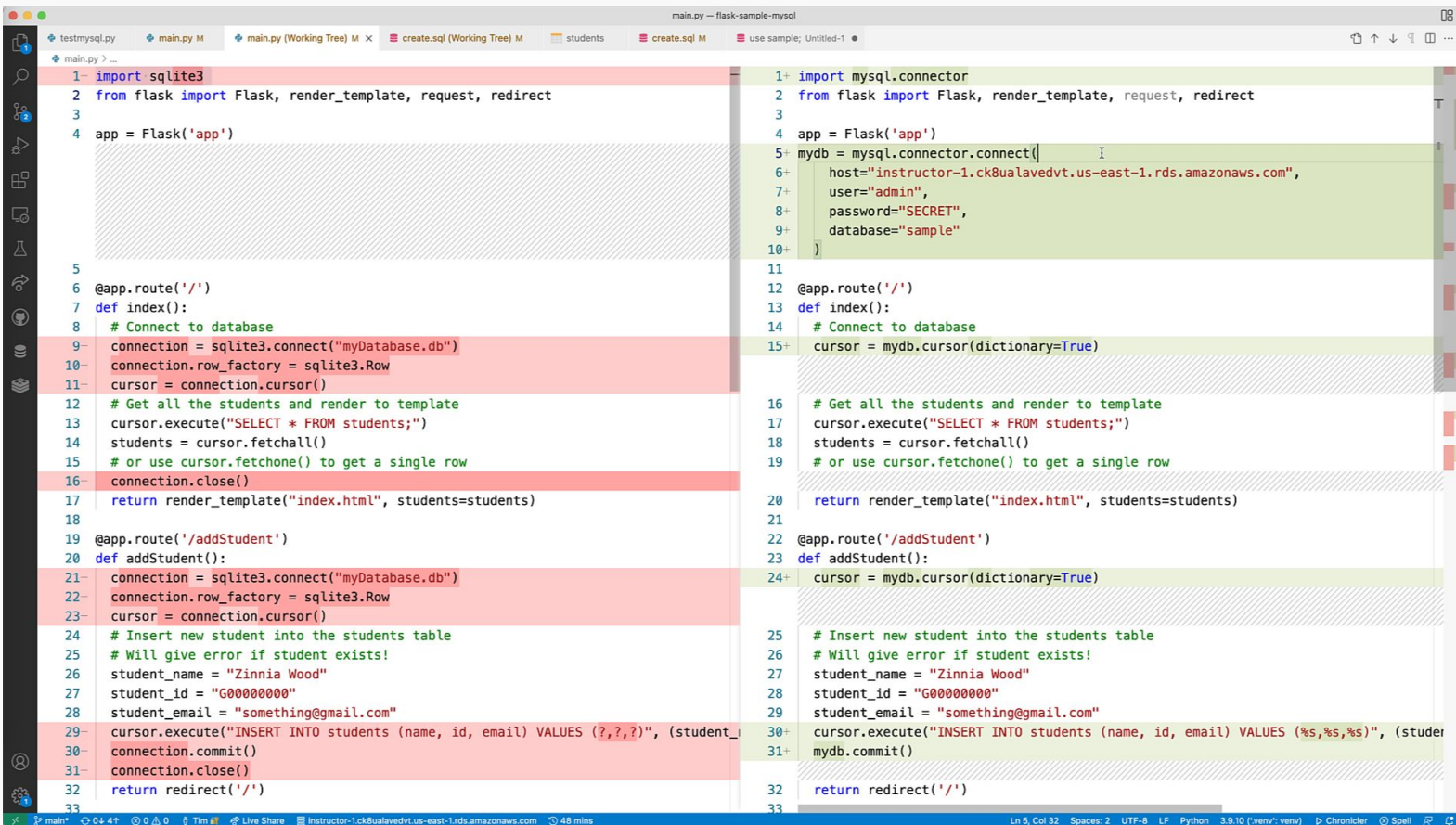
testmysql.py  main.py M  main.py (Working Tree) M  create.sql (Working Tree) M x  students  create.sql M  use sample; Untitled-1 •

create.sql > ...
1- PRAGMA foreign_keys=off;
2
3 DROP TABLE IF EXISTS students;
4 CREATE TABLE students (
5     id      varchar(32) not null PRIMARY KEY,
6     name    varchar(50) not null,
7     email   varchar(50) not null
8 );
9-
10- PRAGMA foreign_keys=on;
11
12 INSERT INTO students VALUES ('G12345678', 'Jett Jacobs', 'jacobsemail@fakeemail.com');
13 INSERT INTO students VALUES ('G22489071', 'Alex Coleman', 'alexcoleman@fakeemail.com');
14 INSERT INTO students VALUES ('G82915273', 'Ethan Baron', 'ethan@fakeemail.com');
15 INSERT INTO students VALUES ('G22004676', 'Cat Meadows', 'cat@fakeemail.com');
16
17 UPDATE students SET name = 'Catherine Meadows' WHERE id = 'G22004676';

1- CREATE DATABASE sample
2-     DEFAULT CHARACTER SET = 'utf8mb4';
3- use sample;
4-
5- SET FOREIGN_KEY_CHECKS=0;
6
7 DROP TABLE IF EXISTS students;
8 CREATE TABLE students (
9     id      varchar(32) not null PRIMARY KEY,
10    name    varchar(50) not null,
11    email   varchar(50) not null
12 );
13- SET FOREIGN_KEY_CHECKS=1;
14
15 INSERT INTO students VALUES ('G12345678', 'Jett Jacobs', 'jacobsemail@fakeemail.com');
16 INSERT INTO students VALUES ('G22489071', 'Alex Coleman', 'alexcoleman@fakeemail.com');
17 INSERT INTO students VALUES ('G82915273', 'Ethan Baron', 'ethan@fakeemail.com');
18 INSERT INTO students VALUES ('G22004676', 'Cat Meadows', 'cat@fakeemail.com');
19
20 UPDATE students SET name = 'Catherine Meadows' WHERE id = 'G22004676';

Ln 9, Col 24 (11 selected)  Spaces: 2  UTF-8  LF  SQL  ▷ Chronieler  © Spell  ↻
```

Python Code Differences



```
main.py — flask-sample-mysql

testmysql.py  main.py M  main.py (Working Tree) M x  create.sql (Working Tree) M  students  create.sql M  use sample; Untitled-1 •

main.py > ...
1- import sqlite3
2 from flask import Flask, render_template, request, redirect
3
4 app = Flask('app')
5
6 @app.route('/')
7 def index():
8     # Connect to database
9     connection = sqlite3.connect("myDatabase.db")
10    connection.row_factory = sqlite3.Row
11    cursor = connection.cursor()
12    # Get all the students and render to template
13    cursor.execute("SELECT * FROM students;")
14    students = cursor.fetchall()
15    # or use cursor.fetchone() to get a single row
16    connection.close()
17    return render_template("index.html", students=students)
18
19 @app.route('/addStudent')
20 def addStudent():
21    connection = sqlite3.connect("myDatabase.db")
22    connection.row_factory = sqlite3.Row
23    cursor = connection.cursor()
24    # Insert new student into the students table
25    # Will give error if student exists!
26    student_name = "Zinnia Wood"
27    student_id = "G00000000"
28    student_email = "something@gmail.com"
29    cursor.execute("INSERT INTO students (name, id, email) VALUES (?, ?, ?)", (student_
30    connection.commit()
31    connection.close()
32    return redirect('/')
33

1- import mysql.connector
2 from flask import Flask, render_template, request, redirect
3
4 app = Flask('app')
5 mydb = mysql.connector.connect(
6     host="instructor-1.ck8ualavedvt.us-east-1.rds.amazonaws.com",
7     user="admin",
8     password="SECRET",
9     database="sample"
10 )
11
12 @app.route('/')
13 def index():
14     # Connect to database
15     cursor = mydb.cursor(dictionary=True)
16     # Get all the students and render to template
17     cursor.execute("SELECT * FROM students;")
18     students = cursor.fetchall()
19     # or use cursor.fetchone() to get a single row
20     return render_template("index.html", students=students)
21
22 @app.route('/addStudent')
23 def addStudent():
24     cursor = mydb.cursor(dictionary=True)
25     # Insert new student into the students table
26     # Will give error if student exists!
27     student_name = "Zinnia Wood"
28     student_id = "G00000000"
29     student_email = "something@gmail.com"
30     cursor.execute("INSERT INTO students (name, id, email) VALUES (%s,%s,%s)", (stude
31     mydb.commit()
32     return redirect('/')
33
```

MySQL vs SQLite Summary

MySQL allows you to run a database on a separate server than your application

—This helps with performance, scalability, and reliability

SQL query syntax is mostly the same

—Remember to **"use"** a database

Python code is mostly the same

—Create one MySQL connection in global scope, then create cursors for each route

—use **%s** to fill in query parameters instead of **?**

Python Virtual Environments

And installing Flask Library



What is a Python Virtual Environment ?

A virtual environment in Python is an isolated environment on your computer!

Allows you to manage project-specific dependencies without interfering with other projects or the original Python installation.

Think of a virtual environment as a separate container for each Python project. Each container:

- Has its own Python interpreter
- Has its own set of installed packages
- Is isolated from other virtual environments
- Can have different versions of the same package

Good for team projects when your teammates have different computers than you, and you don't need to worry about conflicting versions of packages

Extra Resources for Virtual Environments: https://www.w3schools.com/python/python_virtualenv.asp

Virtual Environment

Create environment

```
python3 -m venv .venv
```

Tell VS Code to use the environment when it shows a popup!

Activate environment

```
source .venv/bin/activate    (mac)
.venv\Scripts\activate      (windows)
```

(LATER, after installing all modules)

Save required package list

```
pip3 freeze > requirements.txt
```

Install from required package list (on another machine)

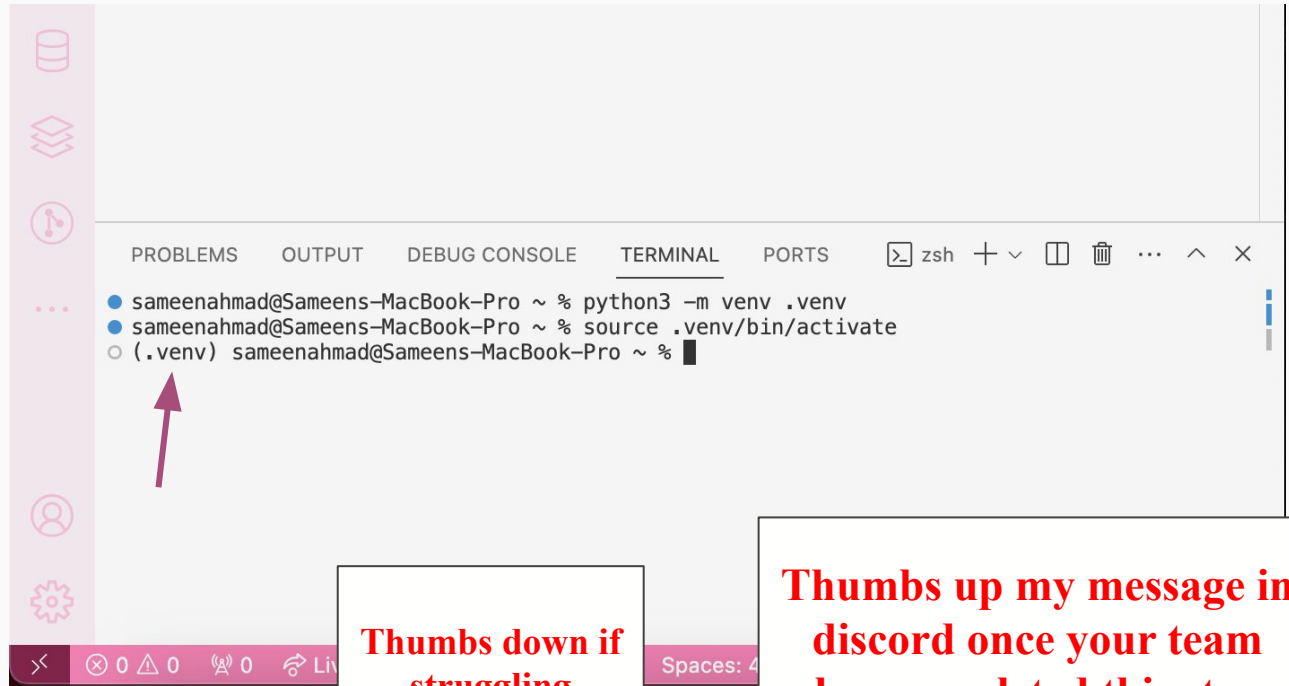
```
pip install -r requirements.txt
```

Use this to save a list of packages you have installed. Then add the file to your git repository

On another computer you could use this to load all the libraries needed by the repository

Check

Will say "(.venv)"
above every command
prompt in terminal if
working correctly



```
...  
● sameenahmad@Sameens-MacBook-Pro ~ % python3 -m venv .venv  
● sameenahmad@Sameens-MacBook-Pro ~ % source .venv/bin/activate  
○ (.venv) sameenahmad@Sameens-MacBook-Pro ~ %
```

Thumbs down if struggling

Thumbs up my message in discord once your team has completed this step

Flask Module Installation

Inside your virtual environment terminal:

pip3 install flask

MySQL Local Setup

VS Code Extension and Python Library

MySQL Extension for VS Code

mysql



MySQL 838K ★ 3.5
MySQL management tool
Jun Han [Install](#)



MySQL 203ms
Database manager for MySQL/M...
cweijan [Install](#)



MySQL Syntax 335K ★ 3
MySQL syntax highlighting supp...
Jake Bathman [Install](#)



SQLTools MySQL/M... 291K ★ 5
SQLTools MySQL/MariaDB
Matheus Teixeira [Install](#)



mysql-inline-decora... 90K ★ 3
Add color coding to inline MYSQL...
odubuc [Install](#)



MySQL Statement S... 36K ★ 5
Easy mysql statement running wi...
Jared Black [Install](#)



MySQL Autocomplete 7K
MySQL Syntax Autocomplete for ...
nospinozacr [Install](#)



ES7 JavaScript/Nod... 23K ★ 5
Simple extension for Node, javas...
abrahamwilliam007 [Install](#)



MySQL v4.8.6

cweijan | 446,350 | ★★★★★ (138)

Database manager for MySQL/MariaDB, PostgreSQL, SQLite, Redis and ElasticSearch.

[Disable](#) [Uninstall](#) [Refresh](#) [Settings](#)

This extension is enabled globally.

Get the one by cweijan!

[Details](#) [Feature Contributions](#) [Changelog](#) [Runtime Status](#)

Database Client for Visual Studio Code

vscode marketplace v4.8.6 installs 447.38K stars 1.7k rating 4.4/5 (139) license MIT

This project is a database client for VSCode, supports manager **MySQL/MariaDB**, **PostgreSQL**, **SQLite**, **Redis**, **ClickHouse**, **达梦**, and **ElasticSearch**, and works as an **SSH** client, boost your maximum productivity!

Project site: [vscode-database-client](#), [中文文档](#)



Database Client

Click DB icon in left menu, Add Database, fill in your RDS connection info

The screenshot shows the 'Connect Server' dialog in DBeaver. The left sidebar has a database icon circled in pink. The 'Connect' button in the top toolbar is also circled in pink. The 'Host' field is filled with '127.0.0.1' and labeled 'RDS Endpoint'. The 'Username' field is filled with 'root' and labeled 'Admin information that you created earlier'. The 'Database' field is filled with 'Connection d'. The 'Connect Timeout' is set to '5000'. The 'SSH Tunnel' is disabled. The 'Use SSL' is disabled. The 'Hide System Schema' is enabled. The 'Use Connection String' is disabled. The 'Connect' and 'Close' buttons are at the bottom.

Thumbs down if struggling

Click here to add the new database connection

Click on this icon to navigate to this extension

Thumbs up my message in discord once your team has completed this step

RDS Endpoint

Admin information that you created earlier

MySQL Python Module

Inside your virtual environment terminal:

`pip3 install mysql-connector-python`

Now save the packages

(Later, after installing all modules)

Save required package list

`pip3 freeze > requirements.txt`

Use this to save a list of packages you have installed. Then add the file to your git repository

Install from required package list (on another machine)

`pip3 install -r requirements.txt`

On another computer you could use this to load all the libraries needed by the repository